
Bastiaan Quast PhD

Machine Learning Engineer & Researcher

| bastiaanquast.com | bquast@gmail.com | [online resume](#)

Python/ML inference engineer with deep expertise in model internals (MLX, llama.cpp, SFT pipelines) and homomorphic encryption. Author of multiple open-source libraries including `mlx-profiler`, `transformer`, `HomomorphicEncryption`, and `HEtools` across Python, R, Julia, and C. Former ML Researcher at the United Nations (ITU) with a strong background in building production ML systems, secure computation, and economic research.

Experience

Owner | Valiq Security

Zurich, Switzerland

2024 – Present - Training LLMs from scratch, i.a. PyTorch, TensorFlow, nanochat.

ML Engineer | Billtrust

New Jersey, USA

2023 – 2024 - Designed and implemented deep learning and machine learning solutions - Worked with Bayesian statistics, data engineering pipelines, and production ML systems

ML Researcher | United Nations (ITU)

Geneva, Switzerland

2017 – 2023 - Built Homomorphic Encryption library (Python, R, Julia) for secure computation on encrypted data - Developed AWS-based (Kubernetes/Docker) speech-to-text and translation platform with custom SageMaker models - Created Flask web application for QR code generation of ITU domains - Developed Android (Java/Kotlin) and iOS applications

Researcher | United Nations (UNCTAD)

Geneva, Switzerland

2016 – 2017 - Computable General Equilibrium modeling (GTAP)

Research Assistant | The Graduate Institute, University of Geneva

Geneva, Switzerland

2013 – 2014 - Taught weekly Masters' Statistics workshops - Supervised Masters' Research Projects

Recent Open Source Software & Libraries

- **mlx-profiler** (2025) — Operation-level profiler for Apple Silicon / MLX (PyPI)
- **HEtools** (2023) — Homomorphic Encryption (BFV, BGV, CKKS) — PyPI & Julia Registry
- **HomomorphicEncryption** (2023) — CRAN package for FHE (BFV, BGV, CKKS)
- **transformer** (2019) — Transformer blocks in native R (CRAN)
- **rnn** (2013) — Recurrent Neural Network architectures (LSTM, GRU) in native R (CRAN)
- **autoresearch-mesa** — Agent harness for automated hyperparameter sweep research in agent-based modeling e.g. population simulations, epidemiology ec.

Education

PhD in Economics

University of Geneva, Switzerland - 2012 – 2016

Master in Quantitative Economics

University of St. Gallen, Switzerland - 2010 – 2012

BSc Economics & BA Theoretical Philosophy

University of Groningen, The Netherlands - 2006 – 2010

Skills

- **Languages:** Python, JavaScript/TypeScript, C/ObjC, SQL, CUDA, Bash, R, Rust, Go, Metal
- **ML / AI:** PyTorch, TensorFlow, MLX, llama.cpp, SFT pipelines
- **Cryptography:** Homomorphic Encryption (BFV, BGV, CKKS), Error-Correction Codes, Encoding, Information Theory
- **Languages:** English (Native), Dutch (Native), Spanish, (Swiss) German, French